

Reproduction of EpiModX, Comparison of LLM+MoE, CNN+MoE, and CNN+BLSTM Architectures for Histone Modifications Prediction in AD and AML Datasets

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1. **Background**
2. **Model/Methods**
3. **Reproduction**
4. **Critiques/Questions**
5. **Investigation and Extensions**

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Biological Background

- Epigenetic modifications contribute significantly to the development of various diseases such as Alzheimer's Disease (AD) and Acute Myeloid Leukemia (AML)
- ~45% of AML patients lack visible chromosomal abnormalities, but defects at molecular level
 - significant portion of AML cases are largely the result of epigenetic factors
- Variations occur in the non-coding region of the genome, making it hard to determine how specifically they contribute to disease states

How can we model the effects of disease-associated histone modifications on disease progression?

Previous Models

- Variant effect prediction models: predict the effects of sequence variants on a variety of chromatin features but not designed to be disease-conditioned
 - DeepSEA and ExPecto
- Histone-specific predictions: identify post-translational modification sites in histone tails but not predictive across disease contexts
 - DeepHistone and Histone-Net

EpiModX

- LLM + CNN + Mixture of Experts
- LLM: learn genomic grammatical structures, syntactic patterns, and semantic relationships from sequence data
- CNN: detect and compress local patterns from LLM representation
- MoE: capture disease-specific epigenetic variations in addition to the general rules of gene regulation

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DNA Language Model

- The backbone of the EpiModX architecture is the DNA Language Model (Caduceus)
- Unlike standard LMs, Caduceus is designed to handle properties of genetic code — bidirectionality, reverse complement symmetry, etc.
- Based on Mamba Selective State-Space Model (SSM), allowing it to scale linearly with sequence length
- Long range, bidirectional, and RC Equivariant
- Produces 16-dimensional numerical feature vector from raw DNA sequence

Caduceus highlights



- ✓ Memory-efficient, bi-directional extension of Mamba
- ✓ RC-equivariant language modeling
- ✓ Improved performance over much larger Transformers-based and comparably-sized SSM-based models

Convolutional Blocks

- Deep neural network module with 4 CNN layers + max-pooling layers was used to reduce spacial dimensions and aggregate information
- Input is processed by 4-block convolutional encoder that progressively extracts hierarchical sequence motifs
 - Each block consists of a 1D convolution with kernel size 5 and same-padding, followed by a ReLU activation and a max-pooling operation with stride 2 that halves the sequence length.
- After 4 pooling operations, the 4096bp input is reduced to a 256-position sequence of 128D feature vectors.

Mixture of Experts

- MoE used to identify the difference between disease progression groups
- Contains 2 MoE attention blocks, customized to capture disease-specific information, and 2 MoE MLP block
- Each MoE block contains 16 experts and 5 disease-specific routing networks for each one of the (5) disease groups
- The output of the MoE attention layer is made up of the noisy routing network and the expert attention head's disease-specific embedding

$$y = \sum_{k=1}^N G_i^k(x) \cdot E_k(x + e_i)$$

- Where i is the disease group, G_i^k is the Noisy Routing network, e_i represents the disease-specific embedding, and E_k denotes the expert (the attention head). Finally, the model had five specialized output heads for each disease group.

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Reproduction Process

Training

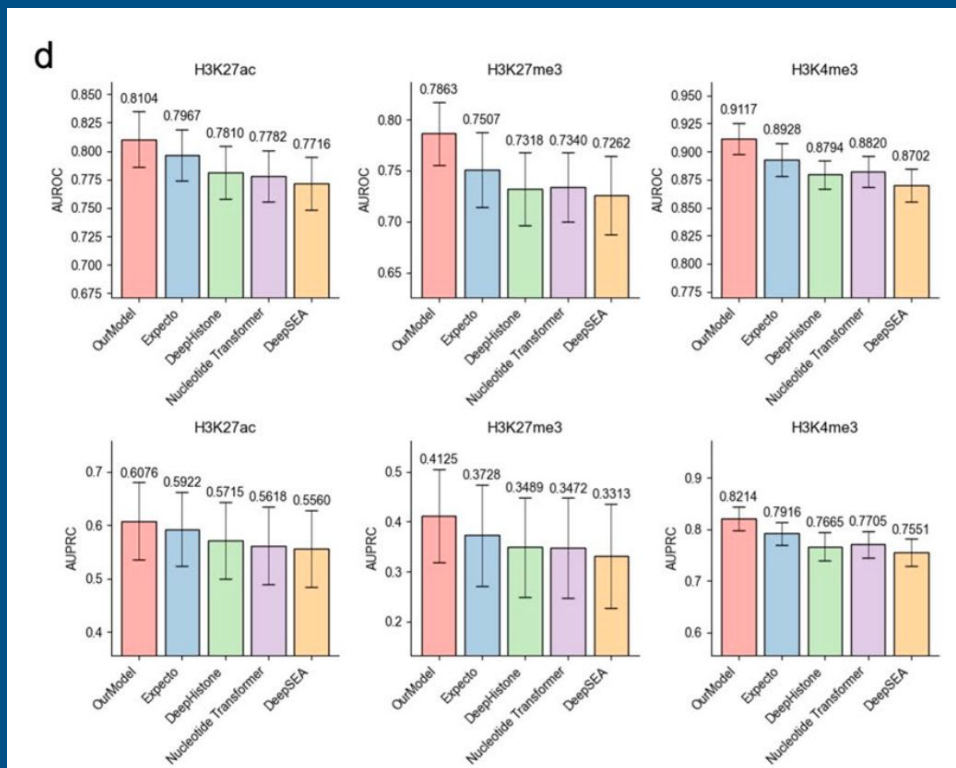
- Model training with data parallelism due to limited GPU capability

Testing

- Resolving checkpoint keys

Successful Reproduction of LLM + MoE Model Training on ALL 3 Histone Modifications in AD Dataset

Original Data in the Paper: AUC and PRC

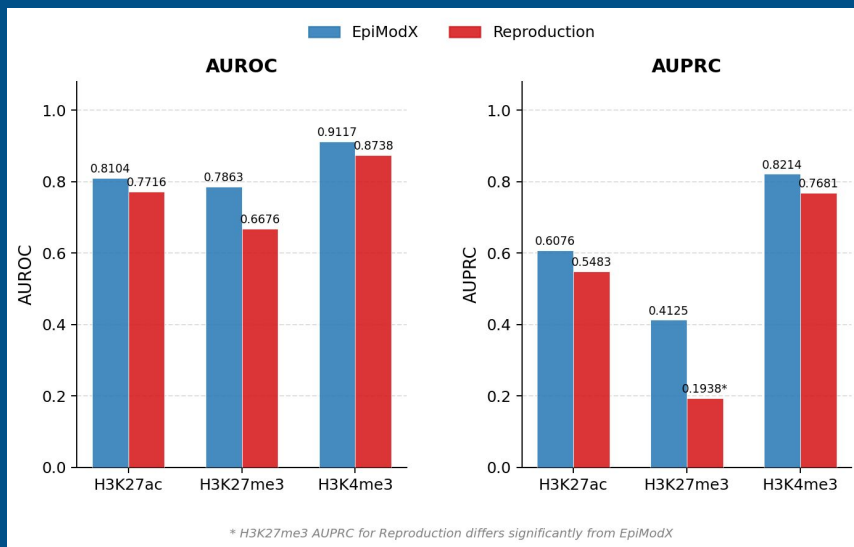


AUROC:
 H3K27ac 0.8104
 H3K27me3 0.7863
 H3K4me3 0.9117

AUPRC:
 H3K27ac 0.6076
 H3K27me3 0.4125
 H3K4me3 0.8214

Reproduced Model Generates Similar Values and Tendency

	EpiModX	Reproduction
AUROC:		
H3K27ac	0.8104	0.7716
H3K27me3	0.7863	0.6676
H3K4me3	0.9117	0.8738
AUPRC:		
H3K27ac	0.6076	0.5483
H3K27me3	0.4125	0.1938*
H3K4me3	0.8214	0.7681

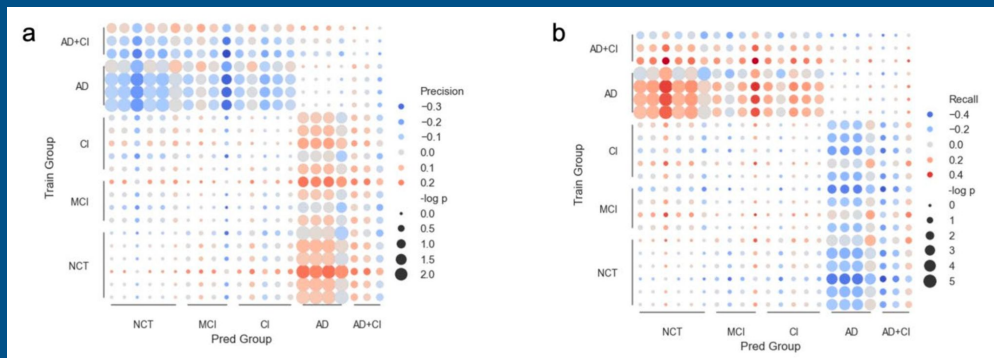


AUROC and AUPRC values:

1. Closely match those reported in the original study.
 2. Consistently preserve the same relative performance trends across the three histone modifications.
- > The reproduced models successfully conduct the class prediction.

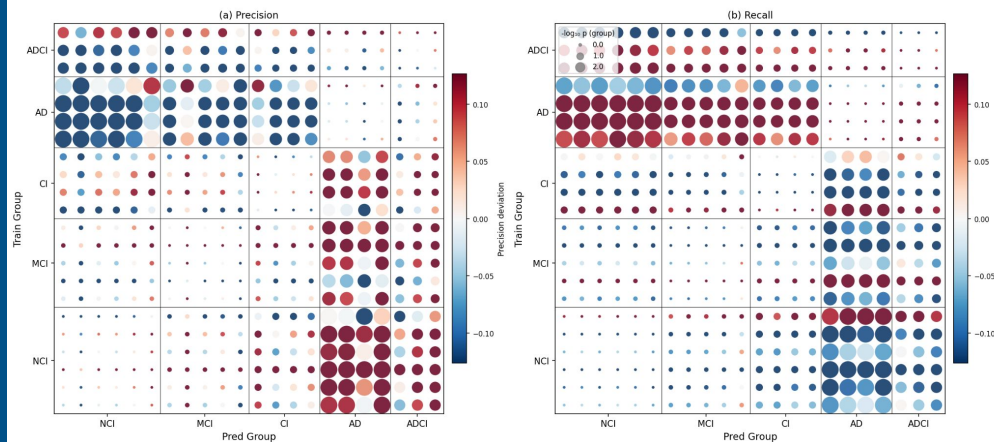
Reproduction Model Generates Similar Prediction Patterns – H3K27ac

Original



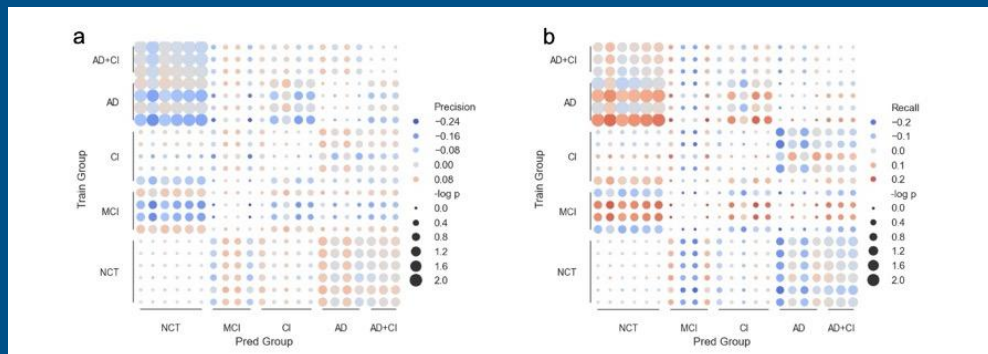
Cross-disease Precision & Recall — LLM+MoE H3K27ac

Reproduced



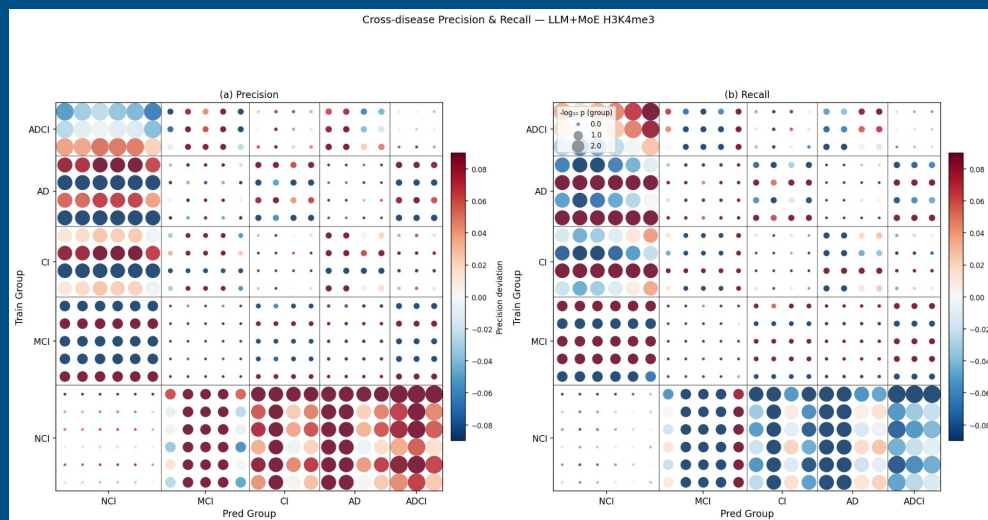
Reproduction Model Generates slightly different Prediction Patterns – H3K4me3

Original



“Parallel analyses of H3K4me3 modifications showed similar inter-group (between the NCI and AD groups) differences.”

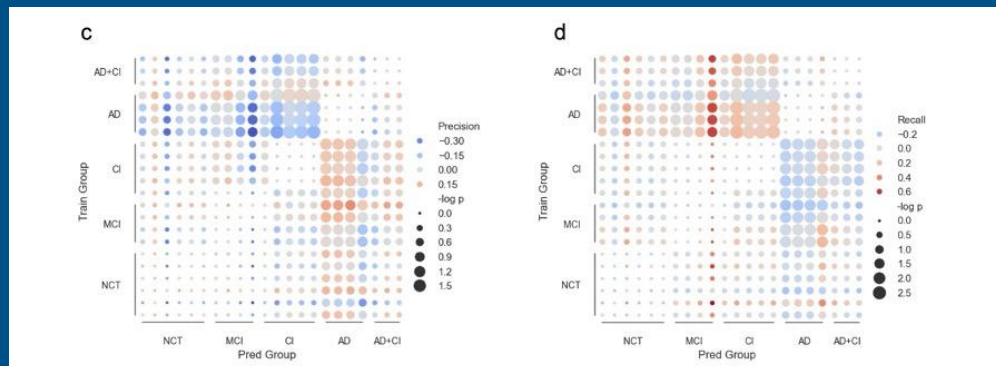
Reproduced



NCI–CI shows less similarity in our reproduction models.

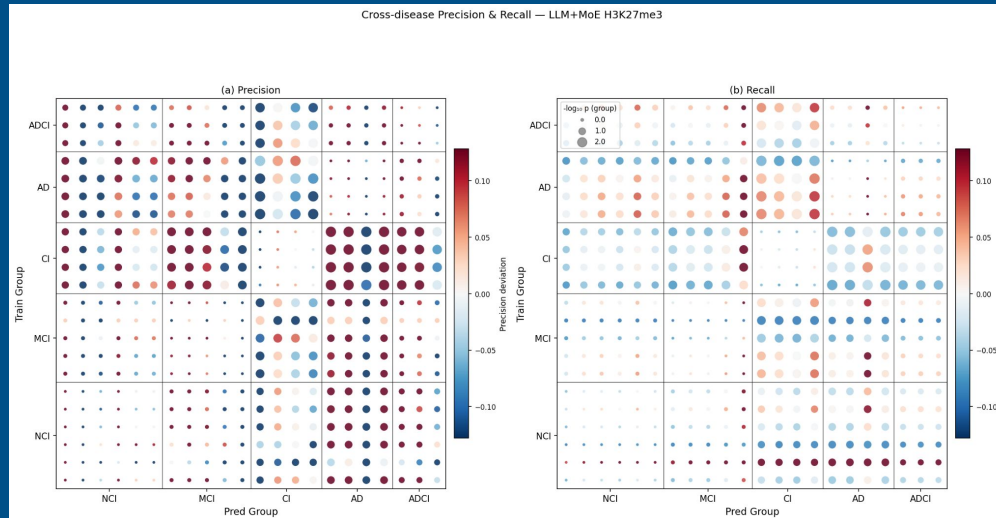
Reproduction Model Generates Similar Prediction Patterns – H3K27me3

Original



“H3K27me3 patterns remained stable across disease states, consistent with its role as a repressive mark less susceptible to disease-associated fluctuations.”

Reproduced



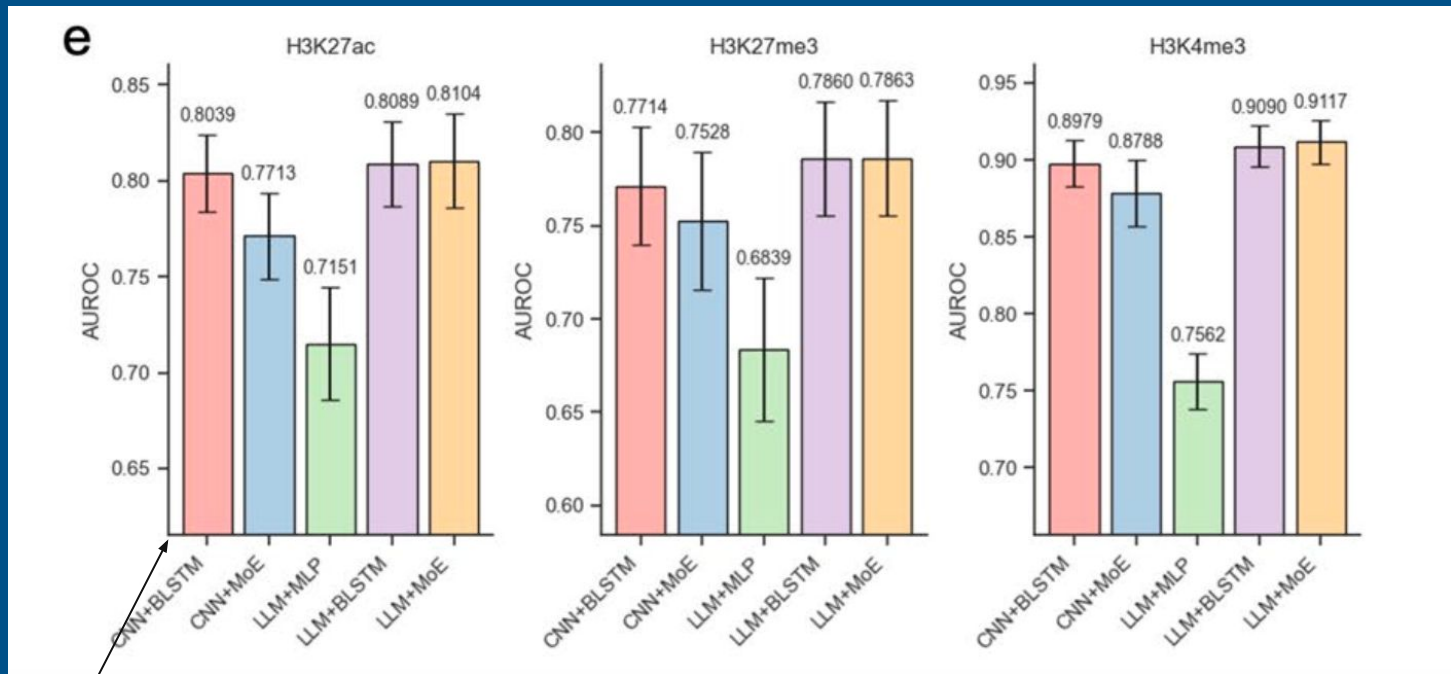
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Critiques

- Expensive architecture → LLM and MoE are both computationally expensive, need to consider tradeoff between actual contribution of models in prediction strength vs. cost
 - We conduct ablation study
- Generalization → paper does not investigate the ability of EpiModX to generalize and be applied to other disease contexts outside of Alzheimer's Disease
 - We introduce AML dataset

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Original Comparison in the Paper: AUC



Even zoomed in on Y axis to make the differences look more significant.

The AUCs aren't different that much.

Q1: Is it worthy to use expensive LLM & MoE to just get small progress?

Q2: Is the small progress even true?

CNN+MoE

- We conducted a partial ablation study with just the CNN+MoE components to compare differences in prediction when incorporating/removing the DNA language model
- CNN+MoE ablation used to identify whether disease-specific performance comes from the MoE mechanism or the pretrained language model
- CNN and MoE architecture stays the same; but the CNN now acts as a traditional feature extractor from raw DNA sequence
 - MoE preserves the ability of the model to leverage disease-specific experts and retain epigenetic differences across various contexts.

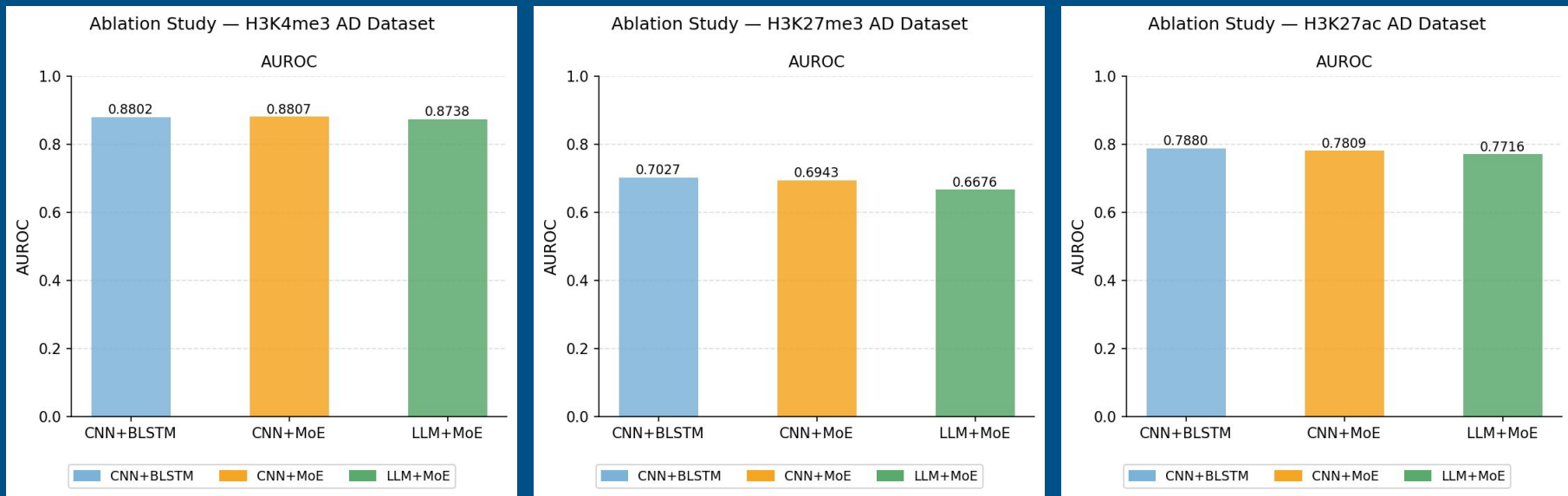
CNN+BLSTM

- Chose CNN+BLSTM because performance ranked 3rd in the paper — slightly lower than LLM+MoE and LLM+BLSTM
 - By contrast, training should be cheaper without LLM blocks
- Another layer of comparison to determine if MoE is essential for learning
- CNN block is identical to CNN+MoE and full model architecture's CNN
- The BLSTM network has a hidden size of 64 per-direction, yielding a 128-dim. hidden state at each position.
 - Dropout = 0,3 between 2 layers
- The recurrent encoder produces a context-aware sequence of 256*128 hidden states
- A global mean-pooling operation collapses the 256-position hidden state sequence into a single 126-dimensional vector summarizing the entire window. This vector is passed through a layerNorm followed by a single linear projection to produce 22 output logits, one per patient.

$$L = -\frac{1}{NP} \sum_{n=1}^N \sum_{p=1}^P [y_{n,p} \log \hat{y}_{n,p} + (1 - y_{n,p}) \log(1 - \hat{y}_{n,p})]$$

Compare the Training Results of LLM+CNN+MoE, CNN+MoE, CNN+BLSTM Model Architectures in AD Dataset

Minimal Global AUC Differences Among Reproduced Models



AUROC values show minimal variation among three model architectures within each histone modifications.

-> the performance gap between EpiModX and baseline models may not be stable.

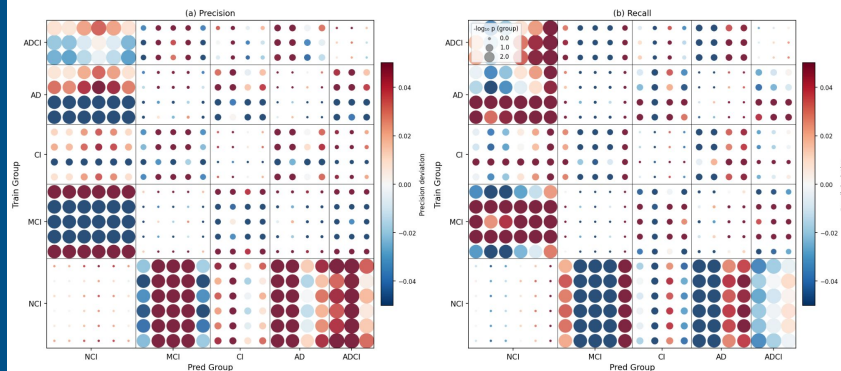
LLM+MoE consistently achieves the lowest AUROC of the three architectures across all three histone modifications, contrary to expectations given its additional complexity.

Q: Whether the architectural complexity of the full EpiModX pipeline translates to a meaningful and reliable performance advantage?

Slightly Different Prediction Patterns Among 3 Reproduced Models – H3K4me3

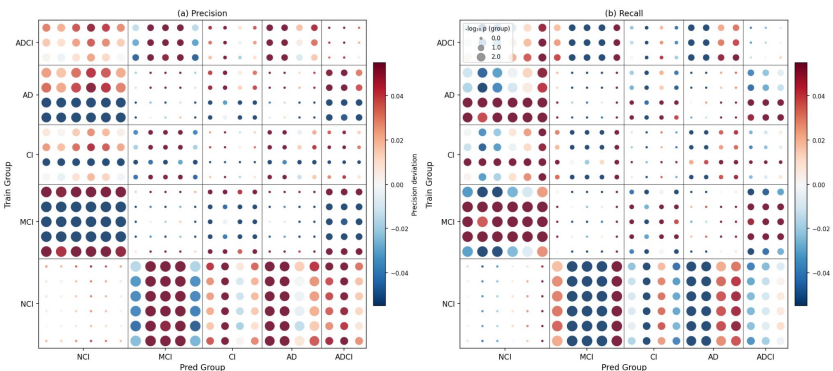
Cross-disease Precision & Recall — H3K4me3 (Figure 3a-b)

CNN-BLSTM



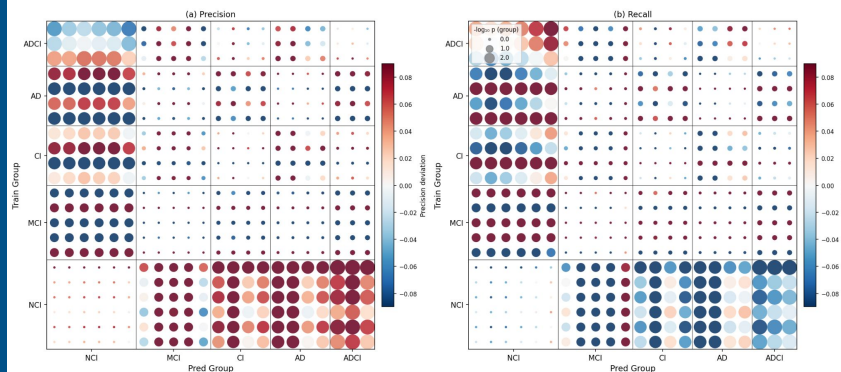
Cross-disease Precision & Recall — CNN+MoE H3K4me3

CNN-MoE



Cross-disease Precision & Recall — LLM+MoE H3K4me3

LLM-MoE

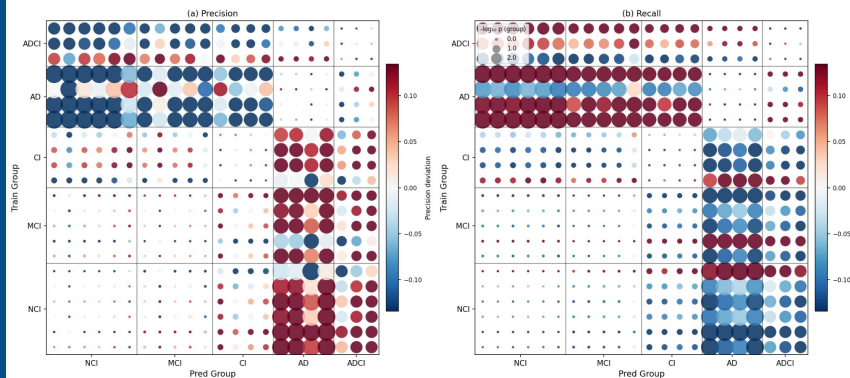


All three model architectures produce consistent cross-patient transfer patterns. The NCI group shows the greatest precision asymmetry relative to MCI, CI, AD, and ADCI. Beyond the NCI separation, inter-group differences are small

Similar Prediction Patterns Among 3 Reproduced Models – H3K27ac

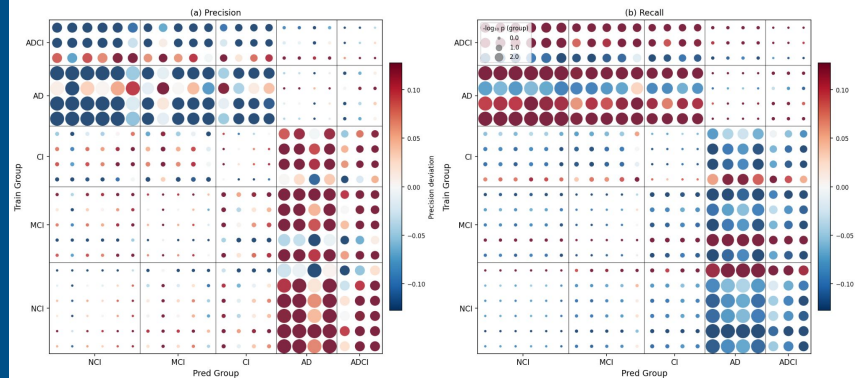
Cross-disease Precision & Recall — H3K27ac (Figure 3a-b)

CNN-BLSTM



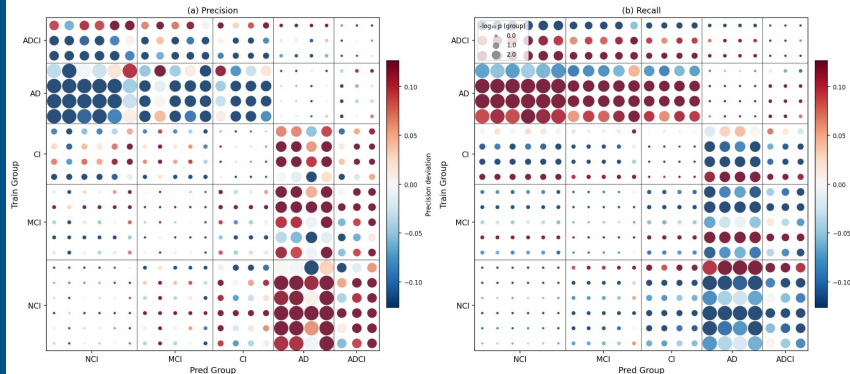
Cross-disease Precision & Recall — CNN+MoE H3K27ac

CNN-MoE



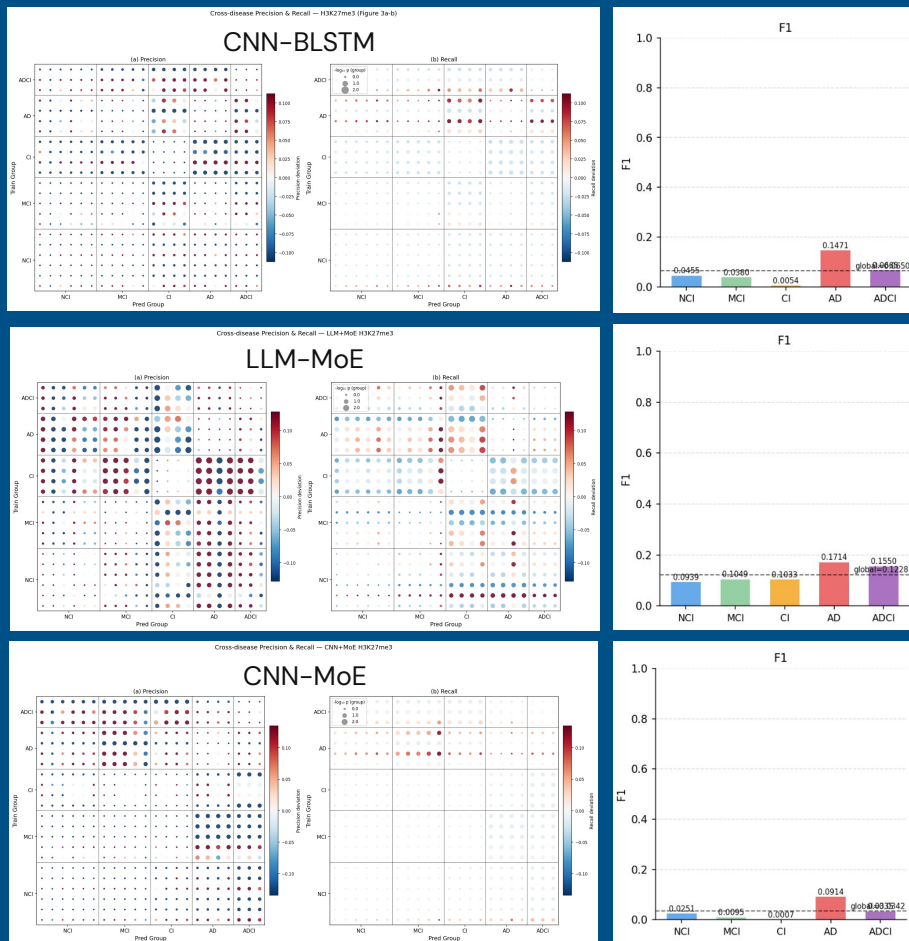
Cross-disease Precision & Recall — LLM+MoE H3K27ac

LLM-MoE



All three model architectures produce more consistent, and even identical cross-patient transfer patterns.

Similar Prediction Patterns Among 3 Reproduced Models – H3K27me3



All show F1 Collapse.

Reasonable!

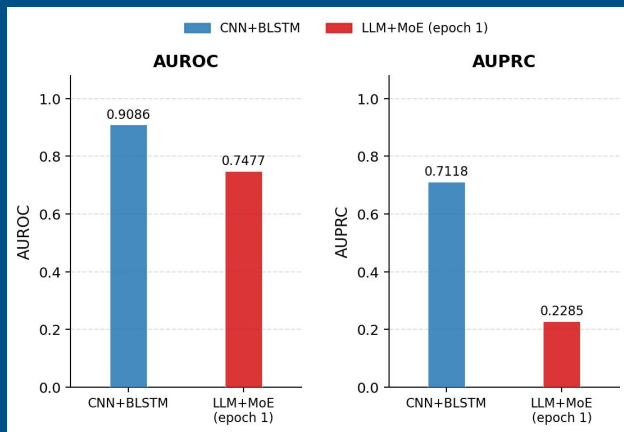
“H3K27me3 patterns remained stable across disease states”!

The inter-group patterns are similar between these three models except that AD, ACI groups are not as well-distinguished from NCI, MCI groups in CNN+BLSTM as in LLM+MoE.

The marginal advantage suggests the LLM backbone extracts weak but meaningful signals that CNN misses.

Generalization to AML Dataset, H3K4me3 Modification

AML Dataset Generalization Comparison – H3K4me3



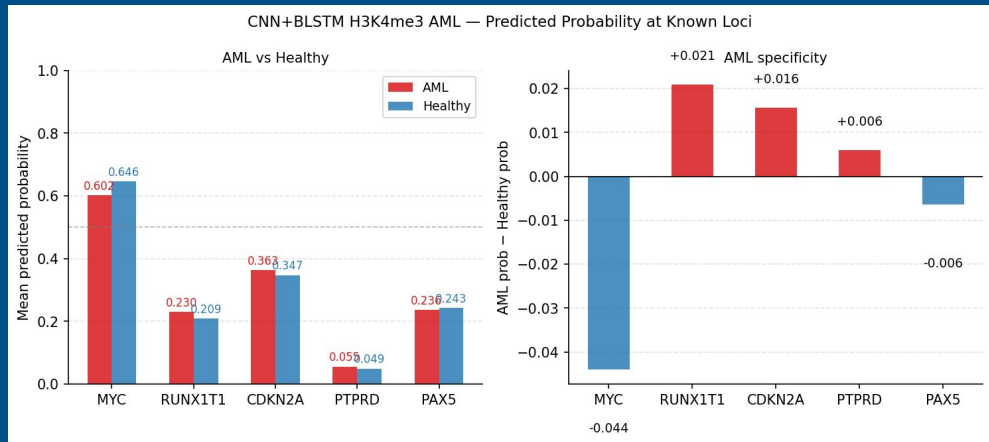
The LLM_MoE model wasn't fully trained due to time limitations.

-> Not a valid comparison between CNN+BLSTM and LLM+MoE_epoch1.

But one question can be asked:

-> Is CNN+BLSTM capturing biologically meaningful signals?

CNN+BLSTM Prediction Show Meaningful Probability Difference at Known Loci



Low Prediction Probability difference:

1. Subpopulation averaging
2. Non-oncogene-specific nature of H3K4me3

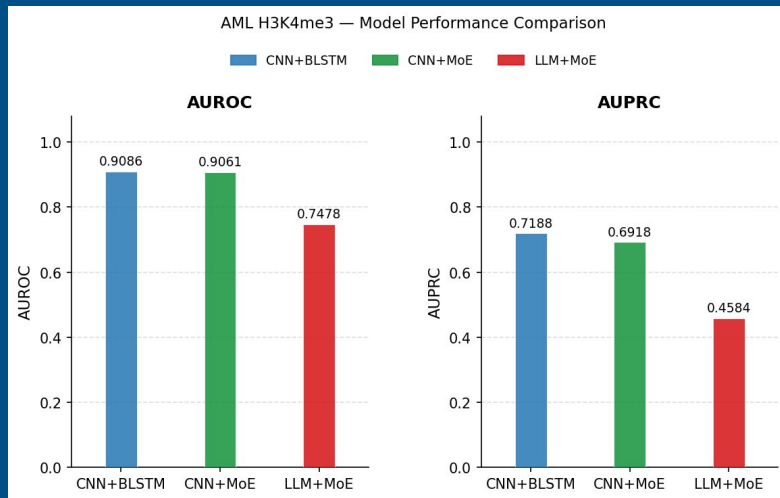
MYC: central proto-oncogene whose promoter marked by H3K4me3 in proliferating cells. Its dysregulation in AML is primarily driven by distal super-enhancer activation, not H3K4me3 modification.

- ↑ RUNX1: hematopoietic master transcription factor and recurrent AML driver. Disrupted in a major AML subtype.
- ↑ CDKN2A: context-dependent regulation in AML through a combination of deletion and retained promoter activity.

PTPRD: tumor suppressor phosphatase frequently inactivated in cancers via copy number loss rather than promoter epigenetic remodeling.

PAX5: B-cell lineage-defining transcription factor, negative control

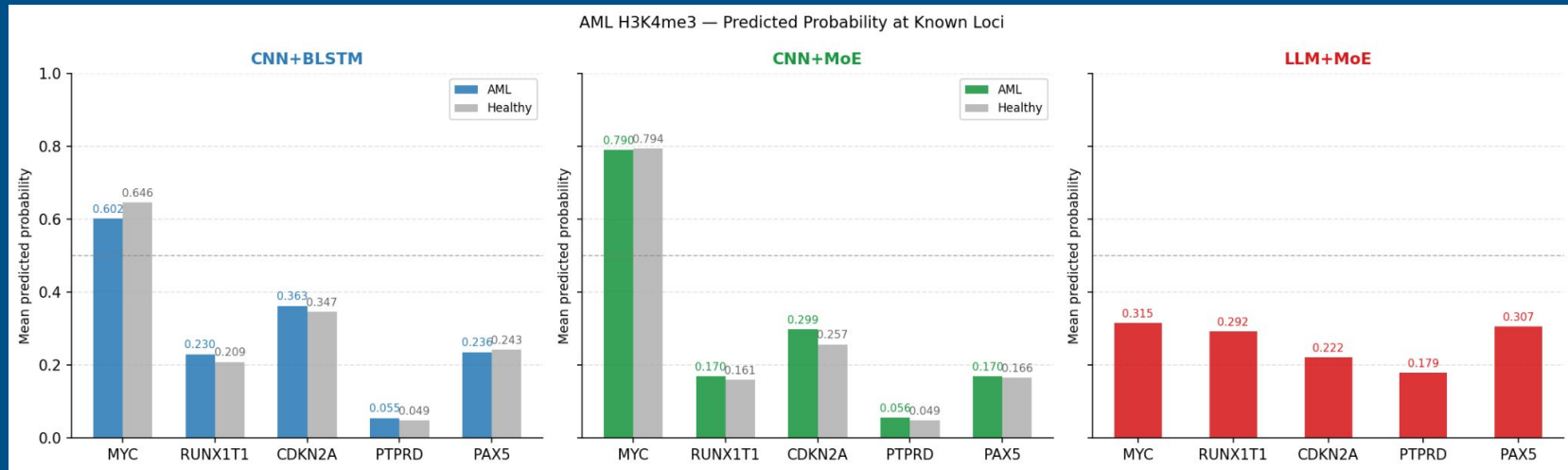
Updated AML Dataset Generalization Comparison – H3K4me3



CNN+BLSTM && CNN+MoE performs well on AML dataset.

* We made a mistake in setting LLM_MoE task as one binary classification instead of two tasks for AML and Healthy. This caused experts collapse and output collapse. Its performance here is not trustable.

Updated Prediction at Known Loci Show similar Prediction – H3K4me3



CNN+MoE model shows similar prediction results to CNN+BLSTM.

*Still, LLM+MoE prediction here is not trustable.

Questions?

Thank You!